Muhammad Hassaan

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SUMMARY

Software Engineer with 3+ years of experience building high-performance backend systems. Improved system throughput by 20% and reduced latency by 15% by optimizing performance-critical C++ modules and multithreaded applications. Experienced in C++, Node.js and Linux, with a strong focus on scalable systems, performance optimization, and backend architecture.

EXPERIENCE

Software Engineer (Backend & Systems)

xFlow Research **December 2024 - Present, Islamabad, Pakistan**

xFlow Research is the Network Infrastructure Services provider company, focused on providing research and development services.

- Improved system throughput by **45%** and reduced latency by **15%** by optimizing performance-critical C++ modules and refining instruction-level execution.
- Increased processing efficiency by **identifying CPU and memory bottlenecks** and implementing low-level performance optimizations on Linux systems.
- Built **multithreaded backend components** to process high-volume real-time data, improving system scalability and reliability.
- Reduced system resource usage by **optimizing memory management and buffer handling** in performance-sensitive modules.
- Improved system stability and deployment reliability by **collaborating with cross-functional teams** and resolving performance-related production issues.

Full Stack Developer (Mobile & Backend)

TechFabbier **September 2023 - November 2024, Remote**

TechFaber offers IT consulting services, specializing in IT infrastructure design and buildout.

- Delivered **weekly production releases** by developing and maintaining mobile and backend features using Flutter and Node.js.
- Improved application performance and user experience by **implementing new features and fixing production issues** across frontend and backend systems.
- Reduced user-reported issues by **debugging and resolving UI, API, and deployment problems** in production environments.
- Built and maintained **GraphQL APIs** to support mobile application features and client requirements.
- Improved deployment reliability by **managing builds, releases, and environment configurations** across multiple environments.

Project Manager & Backend Developer (Intern)

Vialogic **June 2022 - July 2022, Islamabad, Pakistan**

Vialogic offers IT consulting, cloud integration, and analytics solutions for business process optimization.

- Designed and implemented **MySQL database schemas** to support LMS features and improve data management.
- Coordinated development and deployment tasks by **tracking integration steps using Git and Jira**.
- Assisted in backend feature development by **customizing Moodle plugins** and supporting new LMS functionality.
- Ensured successful Linux deployments by **validating builds and resolving integration issues** with cross-functional teams.

PROJECT

Multithreaded Task Processing Server (C++)

Personal Project • December 2024 - Present

- Developed a multithreaded task processing server in C++ to distribute jobs across worker threads.
- Implemented thread synchronization using mutexes, condition variables, and thread-safe queues.
- Improved CPU utilization and reduced task processing time by implementing concurrent task scheduling.
- Designed the system using modular components for task parsing, scheduling, and execution.
- Built the system to run on Linux environments with performance and scalability considerations.

High-Performance Data Processing & Conversion Tool (C++)

Personal Project • December 2024 - Present

- Implemented multithreaded data parsing logic on Linux platforms to accelerate conversion from raw binary files to structured formats.
- Implemented efficient file parsing, buffering, and memory management to handle large datasets.
- Optimized performance by reducing memory usage and improving processing speed through efficient data handling.

- Designed the system to support modular parsing and processing pipelines.

Full Stack E-commerce & Service Platform

Womco • www.womco.co.uk/ • September 2023 - March 2024

- Built a full-stack platform using Flutter, Node.js, and Spring Boot to support customer bookings and service management.
- Designed and implemented RESTful APIs to handle client requests, data processing, and order management.
- Deployed and maintained backend services using AWS (ECS, ECR, S3), improving deployment reliability and scalability.
- Implemented database-driven workflows to manage users, orders, and service data.
- Collaborated in an agile environment using Git and Jira to deliver production features and updates.

Identity & Access Management System Integration

Vialogic • www.keycloak.org/ • June 2022 - July 2022

- Integrated an identity and access management system to centralize authentication across multiple applications.
- Configured MySQL-based user storage and implemented LDAP integration for secure authentication.
- Extended authentication workflows using Java to support custom security requirements.
- Improved system security and access control by implementing centralized authentication and user federation.

Containerized E-commerce Platform

Personal Project • August 2020 - November 2020

- Developed a containerized e-commerce platform using WordPress and Docker.
- Implemented inventory synchronization and multi-channel sales integration.
- Optimized checkout workflow to improve user experience and order processing.
- Managed deployment using Docker and Git for consistent development and production environments.

EDUCATION

Bachelor of Software Engineering

Minor in No Minor • National University of Sciences and Technology (NUST) • Islamabad, Pakistan • 2023 • 3.06

- Relevant coursework: Data Structures, Algorithms, Operating Systems, Computer Networks, Database Systems, Software Engineering, Distributed Systems, Algorithms and Data Structures.

CERTIFICATIONS

Cybersecurity Certified (CC)

$(ISC)^2$ • 2023

- Validates foundational knowledge in cybersecurity, including network security, risk management, access control, and security operations. Supports secure backend development, system hardening, and secure authentication system implementation.

Scrum Fundamentals Certified (SFC)

Scrum Study • 2021

- Demonstrates knowledge of Agile development methodologies and Scrum practices, including sprint planning, task tracking, team collaboration, and iterative software delivery in professional development environments.

SKILLS

Languages: C, C++, Go, JavaScript, Python, Java, SQL

Backend & Systems: Node.js, Express.js, REST APIs, GraphQL, Multithreading, Linux, System Programming, Performance Optimization, Networking, Concurrent Programming

Frontend & Mobile:React.js, Flutter, WordPress, Laravel, HTML, CSS

Databases: MySQL, DynamoDB

Cloud & DevOps: AWS (ECS, ECR, S3, Lambda), Docker, CI/CD, Git

Testing & Tools: JMeter, Selenium, Jira, Postman

Cybersecurity & Networking Tools: Wireshark, Nmap, Burp Suite, Aircrack-ng, Cisco Packet Tracer